- The acceleration due to gravity is denoted by $g\,\mathrm{m\,s^{-2}}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

**12** In this question the unit vectors **i** and **j** are horizontal and vertically upwards respectively.

A particle has mass 2 kg.

**(a)** Write down its weight as a vector. **[1]**

A horizontal force of 3 N in the **i** direction and a force $\mathbf{F} = (-4\mathbf{i} + 12\mathbf{j})\,\mathrm{N}$ act on the particle.

**(b)** Determine the acceleration of the particle. **[3]**

**(c)** The initial velocity of the particle is $5\mathbf{i}\,\mathrm{m\,s^{-1}}$.

Find the velocity of the particle after 4 s. **[2]**

**(d)** Find the extra force that must be applied to the particle for it to move at constant velocity. **[1]**